

# Jayadithya R

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## GENAI ENGINEER, RUST TOKENIZER INFRA PLUS LLM EVAL

New-grad AI engineer shipping bottom-up systems: Rust BPE tokenizer 11-16x faster than SentencePiece, LoRA-tuned 7B beating 14B at 80.39% on vuln eval, and pipelines over 1.18M chunks. MLIR intern grounding in compilers, seeking LLM application and NLP infra roles.

## SKILLS

**Languages:** Rust | Python | TypeScript | SQL

**AI/ML:** LoRA fine-tuning (Unsloth) | Hugging Face Hub | ChromaDB | CLIP/transformers.js | Gemma 4-bit inference

**Backend/Data:** FastAPI | PostgreSQL | SQLite | Meilisearch | Docker | Next.js | MLIR/LLVM IR | GitHub Actions CI | scrapy

## EDUCATION

**B.Tech Artificial Intelligence and Data Science, Dr. M.G.R. Educational and Research Institute, Chennai, TN** | 2026  
GPA: 8.0/10.0

## PROJECTS

**tamil-tokenizer** ([github.com/barryallen16/tamil-tokenizer](https://github.com/barryallen16/tamil-tokenizer))

- Built BPE tokenizer for Tamil in Rust (1,814 LOC, 23 tests), training 32K vocab on 128K Tamil Wikipedia lines and publishing it to Hugging Face Hub
- Benchmarked Aho-Corasick O(n) encoder with rayon-parallel trainer and heap-optimized merges, 11-16x faster than SentencePiece/HF Tokenizers on Tamil

**VulnChecker-Java** ([github.com/barryallen16/VulnChecker-Java](https://github.com/barryallen16/VulnChecker-Java))

- Tuned Qwen2.5-Coder with LoRA (r=64, Unsloth) through 7-stage data pipeline for Java vulnerability detection, publishing datasets to Hugging Face
- Evaluated 7B fine-tune at 80.39% on 102-case harness, beating the 14B base, proving small-model specialization wins here

**sivabharani-comments** ([github.com/barryallen16/sivabharani-comments](https://github.com/barryallen16/sivabharani-comments))

- Engineered pipeline over 2,755 Tamil videos with multiprocessing proxy validation, hash-checkpointed resume, and incremental JSONL appends
- Processed 1,185,665 transcript chunks through local Gemma 4-bit inference and published the resulting dataset to Hugging Face for reuse

**fitcheck-family** ([github.com/barryallen16/fitcheck-website](https://github.com/barryallen16/fitcheck-website))

- Distilled Qwen3-VL-8B teacher into 2B student via black-box pipeline: 106K images to attribute records, 538K caption-QA pairs for LoRA (r=32) tuning, 2.9K rationale-annotated recommendations
- Held 91.6% of teacher quality at 3.14x less VRAM on 1,000-image bench; blind human study (297 ratings) confirms auto-eval tracks human judgment (r=0.303, p<0.05)

**kaggle-automation** ([github.com/barryallen16/kaggle-automation](https://github.com/barryallen16/kaggle-automation))

- Deployed Dockerized FastAPI control plane (compose, GitHub Actions CI) pooling Kaggle API keys to shard GPU jobs across accounts
- Provisioned 12-hour session monitor with Telegram alerts and quota UI across 223 logged runs; 12 test suites guard the scheduler

## WORK EXPERIENCE

**Blubridge Technologies, Chennai, TN | AI Research Intern** | 07/2025 - 10/2025

- Architected operator-fusion passes in MLIR for custom tensor dialect, removing redundant kernel launches to cut launch overhead ~20% on benchmark workloads
- Migrated high-level dialect ops down to LLVM IR with shape inference and bufferization for dynamic tensors, stabilizing codegen for variable-shape models
- Hardened pass correctness with 40+ lit tests covering IR round-trip stability, wired into CI to block regressions